Exercise 5: oxygen in the water column

1. Write the equation for the oxygen dynamics in the water-column, including boundary conditions, and add the oxygen-control switches to the other equations.
2. Run the model for two scenarios: one “standard”, and one with a mixed water column with high diffusivity. How does the oxygen profiles differ, and why?
3. Choose two parameters for a sensitivity analysis. Run the model for different values of the parameters under a standard summer condition. Prepare plots of the profiles of N, P, Z, D, and O that illustrates how the parameter affects the solution.

$k_{O_{surf}} = 10 \ldots 20$ m/day

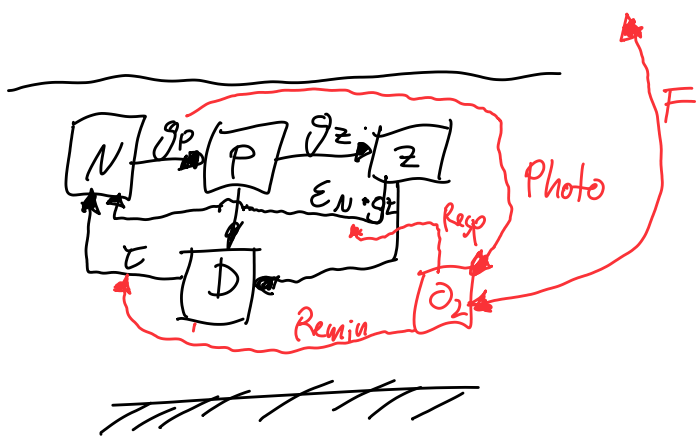
$k_{O_{Bot}} = 1$ m/day

$Y_P = 5$

$Y_N = 6{,}625$

$$\frac{dO_2}{dt} = \text{Diffusion} + \text{Photo} - \text{Resp} - \text{Remin}$$

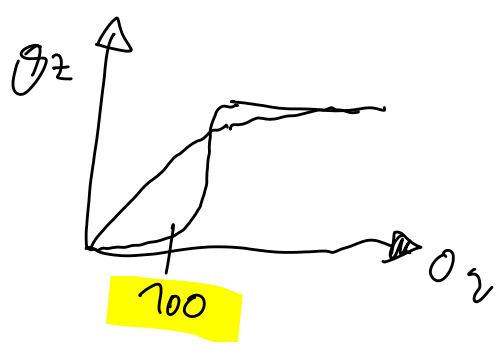
[mmol O_2 m^{-3}]



$\text{Photo} = P \cdot g_P \cdot Y_P$ → yield $\frac{\text{mmol}\,O_2}{\text{mmol}\,N}$

$\text{Resp} = Z \cdot \varepsilon_n \cdot g_Z \cdot Y_N$

$\text{Remin} = \tau \cdot D \cdot Y_N$



$$J_{surf} = -(O_{2eq} - O_{2atm}) \cdot k_{O_{surf}}$$

260

$$J_{bot} = (-O - O_{2,n})\, k_{Bot}$$